

Arrows in Their Backs: Vocabulary Lead and Product Survival in the US Trademark Record

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Abstract

Are pioneers the ones with arrows in their backs? The US trademark record can answer at census scale: a trademark application is a dated description of a product a firm expected to sell, and five years into a registration the owner must swear the product is still in commerce or lose the mark. Applications filed 1995–2019 are scored on how far their goods/services language sat toward where their industry’s language was moving (lead) and on how unusual it was for the industry (atypicality), and each mark and its owner is followed through registration, the five-year proof of continued use, the year-ten renewal, a priced private round, SEC reporting and listing. Across measures of both survival and the likelihood of reaching financial markets, being early costs. Among 3.10 million registrations, the most leading fifth fail the five-year proof 1.4 points more often than the most lagging fifth, within Nice class and registration year with drafting and counsel held ($t = 8.3$), in three of every four classes and under every partition of the vocabulary tried. The penalty was about twice as large for late-1990s filings and returned from 2008. The exception is 2000–2004, when it reversed in three of the four technology classes: marks still written in the computer and information-services vocabulary that had led the late 1990s failed at among the era’s highest rates, while leading filings in goods and retail themes were rewarded, most strongly in online retail. Unusual language behaves differently. Where a theme is new only to a Nice class, arriving from elsewhere in the corpus, its filings survive the proof more often, though less often than equally atypical filings; themes new to the entire corpus are too rare in the record to judge. Registration responds to atypicality, peaking at a middling distance from the class’s usual language, and cannot respond to lead, which does not yet exist when an application is examined; the five-year proof responds to lead. No later rung rewards being early: leading debuts raise priced rounds no more often than lagging ones, and reach SEC reporting and listing less often.

Keywords: pioneering; first-mover disadvantage; optimal distinctiveness; trademarks; product survival; text as data.

JEL: O31, M13, L10, O34.

1 Introduction

Whether market pioneers are advantaged or burdened is an old question that has been answered mostly on named markets and on the firms that survived long enough to be remembered (Golder & Tellis, 1993; Semadeni & Anderson, 2010). The US trademark record offers a different sample. Every firm that intends to sell a product under a mark files a dated description of that product; the description is written by the millions of firms that never patent as well as by those that do; and between the fifth and sixth year of a registration the owner must swear that the product is still

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in commerce or lose the mark. Whether a firm was early can therefore be read from what every entrant wrote, and whether its product lasted can be read from what every owner later swore. This paper asks the pioneer’s question on that record.

Being early is measured from the language itself. A topic model, fitted once on a sample drawn equally from every industry and then applied to every filing, reduces each goods/services description to a mix of themes, and each filing’s theme mix is compared, by Kullback–Leibler divergence, with what its own Nice class filed in the five years before and the five years after it. The average of the two comparisons is the filing’s *atypicality*, how unusual it was for its industry; their difference is its *lead*, positive when the industry’s language subsequently moved toward the filing’s. Lead is computed from filings not yet made, so no one at the time could observe it; it is a positional quantity, not a causal one. The construction, its nomenclature, and the validation program that supports it are the subject of a companion paper (Silver, 2026), and Section 3 summarizes what a reader of this one needs.

The answer is yes. Among 3.10 million registrations issued 2002–2018, marks whose language most led their industry were cancelled for non-use at the five-year proof 1.4 points more often than marks whose language most lagged it, within Nice class and registration year, with drafting and counsel held and errors clustered on the owner ($t = 8.3$); the contrast with fixed effects alone is +2.3 points, positive in 33 of 44 classes, and it holds under every partition of the vocabulary tried. Being unusual carries no such penalty. Its apparent protection at the proof mostly disappears when themes are fitted within each class, and filings that carry a theme new to their class, imported from elsewhere in the corpus, survive more often than established ones but less often than equally atypical filings.

The penalty is not constant, and where it moved is informative. It was about twice as large for marks filed in the late 1990s as for any cohort since, reversed for technology marks filed 2000–2004, and returned from 2008; the swing is largest inside the four technology classes, while outside them the penalty stayed positive in every era. Much of the swing is between themes rather than within them: each era’s leading vocabulary was concentrated in one or two themes — computer and information services in the 1990s, downloadable software after 2008 — that failed at their own rates. In 2000–2004 the pattern turned over: the information-services vocabulary had left the leading fifth and its marks failed at among the era’s highest rates, while leading filings in goods and retail themes were rewarded, most strongly in online retail. Web-based filings fail hardest in each class’s own first internet years and converge toward their class within a decade. Joining a surging theme costs little on average, and arriving early within a wave costs nothing further.

The two steps the record dates respond to different things. Registration — mostly a measure of the applicant’s own follow-through — rises with atypicality and turns down only at the extreme, and barely moves with lead, which did not yet exist to be read. The five-year proof, recorded after the industry’s language has moved, penalizes having been early. Following the owners further, no rung of financial success rewards it: leading debuts raise priced rounds no more often than lagging ones and reach SEC reporting and listing less often. The paper also records what applicants learn: those who refile an abandoned application rewrite toward the class’s typical description, from both tails, whoever drafts the second attempt.

Section 2 places the exercise against the pioneering, optimal-distinctiveness and trademark-indicator literatures. Section 3 states the measure and the record’s steps. Sections 4–7 follow the mark and its owner through registration, the five-year proof, the waves that moved the penalty, and the capital markets. Section 8 reads the stages together, and Section 9 summarizes what changes under other choices.

2 Pioneering, conformity, and the trademark record

2.1 Pioneering and optimal distinctiveness

The pioneering literature has measured first-mover advantage mostly on named markets and survivor-prone samples (Golder & Tellis, 1993; Semadeni & Anderson, 2010). A parallel literature finds evaluators reward conformity with a penalty for extremity in both directions — the optimal distinctiveness of Zuckerman (1999), Hsu (2006), Zhao et al. (2017), and Taeuscher & Rothe (2021). This paper meets both at census scale: every entrant’s language is observed, whether or not the product survived to be remembered, and the record dates two steps that respond to the two components differently. Registration peaks at a middling distance from the class’s usual language and cannot observe lead; the five-year proof of continued use penalizes having been early.

2.2 Trademarks as innovation data

A trademark filing is a dated, costly, public disclosure made by the millions of firms that never patent, and a literature has grown around reading it as an innovation indicator (Castaldi, 2020; Willeke et al., 2026). Trademark counts and breadth predict venture valuations (Block et al., 2014); filing timing tracks innovation processes (Seip et al., 2018); and trademarks mark the product-market side of innovation that patents miss (Flikkema et al., 2019, 2025). Census-linked work reads registration as a correlate of firm growth and survival (Dinlersoz et al., 2018, 2023). Nearly all of it counts filings, classes, or renewals; the descriptions’ text enters, if at all, as breadth or similarity. This paper reads the text itself, and reads it against time.

2.3 Text measures of novelty

Scoring a document by its divergence from a reference corpus before and after it comes from cultural analytics: Murdock et al. (2017) on Darwin’s reading notebooks, Barron et al. (2018) on the French Revolution’s parliamentary debates. In economics and finance, text-based measures locate firms in product space (Hoberg & Phillips, 2016), date technological innovation in patents over the long run (Kelly et al., 2021), and track the vintage of patent language (Kalyani, 2024). The construction here differs in its object — the composition of commercial offerings, not ideas or claims — and in its two-sided reference: every filing is scored against its own industry’s language both before and after its date, which is what lets *unusual* be separated from *early*. The separation matters because the two components predict opposite things.

2.4 Nomenclature

The innovation literature is not consistent about what quantities like these are called, and several of its established terms assert things this design cannot show. *Novelty* and *transience* (Murdock et al., 2017) name the two KL levels themselves, and novelty is declined because it names only the backward-looking half of a quantity that faces both ways; *resonance* (Barron et al., 2018) is arithmetically what this paper calls lead and is avoided because it names a mechanism the design cannot observe; *radicalness*, *originality* and *generality* (Hall et al., 2001; Verhoeven et al., 2025) each claim something about the product or its realized influence that a description’s position cannot carry. The companion paper gives the full argument for each term and a table mapping the five quantities to their nearest occupants (Silver, 2026).

3 The measure and the record in brief

3.1 Two readers, one living backwards

Imagine two readers of a filing’s goods/services description, each having seen a disjoint half of its industry. One has read only what the industry filed in the five years before it. The other has read only the five years after. Each is asked how surprising the wording is. The measurement is how much they disagree: a description that startles the first reader and not the second arrived before its industry’s language moved that way.

3.2 From a description to two numbers

The surprise is measured on themes rather than on individual words. A theme is a list of words that tend to turn up in the same descriptions — *video, computer, electronic, music, games, say, or store, retail, store services, featuring*. The lists are not written by hand. They are found by latent Dirichlet allocation (Blei et al., 2003), a standard method that starts from a random guess about which words belong together and adjusts the lists, pass after pass, until each description can be explained by a few themes and each theme uses a consistent set of words. The model is fitted once, on 10,000 descriptions from each of the 45 Nice classes, and then applied to every filing, so that each description becomes a mix of themes that sums to one — for example 40% of one theme and 15% of another. We use fifty themes, roughly one per Nice class. That is coarse by design: at fifty every theme’s leading words genuinely occur together, while at 200 and 500 themes a tenth and a fifth of themes do not (Lau et al., 2014); and the findings below do not depend on the choice, since the companion paper rescores every filing at 200 and 500 themes and with fifty themes fitted separately inside each class, and the headline penalty holds under all of them (Silver, 2026).

Write P for a filing’s theme mix, and Q^- and Q^+ for the average theme mix of every same-class filing in the 1,826 days before and after its filing date. The filing’s past-facing and future-facing surprise are the Kullback–Leibler divergences $K^- = \text{KL}(P \parallel Q^-)$ and $K^+ = \text{KL}(P \parallel Q^+)$, in nats: zero when the filing’s mix matches its class’s, and larger the more weight the filing puts where the class puts little. The two axes used throughout are their average and their signed difference:

$$A = \frac{1}{2} (K^- + K^+) \quad (\text{atypicality}), \quad L = K^- - K^+ \quad (\text{lead}).$$

A filing is *leading* when $L > 0$: its language was closer to what the class wrote afterwards than to what it wrote before. The two surprises correlate at 0.99 — a filing far from its class’s past is almost always far from its future too — so their average carries the large shared component, how far from the class the filing sits, and their difference carries the small residual that holds the timing. AMAZON.COM’s 1997 filing in advertising and retail sits at the 47th percentile of its class on atypicality and the 94th on lead: the class moved toward its description over the following five years and had not been writing that way in the five before.

Three properties of the construction matter for reading the results. First, themes rather than words: a divergence measured on words mostly counts description length, because a short description cannot spread over many words, and word-scored atypicality correlates with log distinct terms at -0.64 ; with themes the correlation is between -0.10 and $+0.10$. Second, within class rather than across it: descriptions are drafted from the USPTO’s Identification Manual, house precedent and competitors’ filings, so a class’s spread of atypicality reflects its drafting conventions rather than its inventiveness. Every estimate is taken within class and year, and magnitudes compare across classes only as signs and orderings. Third, the measure reads which themes a filing draws on and how heavily, not how it combines them, so novelty assembled from ordinary parts is close to

invisible to it. Nomenclature follows Murdock et al. (2017) and Barron et al. (2018) in arithmetic and departs from them in names (Section 2.4).

3.3 The record’s steps, the samples, and the conventions

The outcome sections that follow are written to be read independently, so the record’s steps, the outcome names, and the counting conventions are set out once, here.

A US trademark passes through a sequence of dated, high-stakes steps, and each step leaves a record. An application is filed with a goods/services description; an examiner reviews it, often issuing office actions the applicant must answer; if allowed and (for intent-to-use filings) a sworn statement of use is filed — the proof that the mark has entered commerce — the mark registers. Registration starts a maintenance clock: in years five to six the owner must file a Section 8 declaration proving continued use in commerce, with a six-month grace period — the *five-year proof of continued use*, the paper’s central outcome; around year ten a combined Section 8 declaration and Section 9 renewal is due, and every ten years thereafter. A mark that misses a proof is cancelled. Madrid Protocol registrations (§66(a) basis) file Section 71 declarations on the same schedule and are flagged separately throughout. Graham et al. (2013) describe the administrative data; the event-dated corpus built here adds the full prosecution history of each case, so a cancellation is assigned to a specific proof by its date rather than inferred from a status snapshot.¹

Table 1: The trademark funnel in this corpus. Registration counts are unique serials; five-year-proof outcomes are event-dated (a Section 8 cancellation at registration age 4.0–8.5 years). Source: the re-parsed USPTO application backfile, scored as in this section.

Stage	<i>n</i>	Share
Debut filings scored, 1995–2018 (owners’ first filings)	3,589,068	
reached registration	2,066,600	57.6%
All unregistered class-records filed 1995–2018	3,798,318	
never filed statement of use		42.6%
stopped responding in prosecution		45.3%
removed adversarially		4.3%
Registrations 2002–2018, scored, unique serials	3,104,534	
failed the five-year proof of continued use (event-dated)		52.3%
Passed the five-year proof, cohorts 2002–2013, status known	1,129,724	
failed at the year-ten renewal		39.1%

Three named outcomes recur (Table 1). *Failing to register*: the application never completed, which happens by the applicant’s own inaction in roughly nine cases of ten (Section 4). *Failing the five-year proof of continued use*: a registration cancelled for non-use at registration age 4.0 to 8.5 years, which means the product the mark named left commerce within six years (Sections 5 and 6). *Reaching the SEC record*: the owner appears in dated SEC events — a priced Form D round, financial reporting, a listing (Section 7).

Outcomes are rates, so every estimate is a shift in a probability and is stated in percentage points against the base rate it shifts: +1.4pp on a 52.3% base means 53.7% against 52.3%. The basic comparison is the outcome rate of the fifth of filings with the highest lead minus that of the

¹Cross-checked against the USPTO status-code dictionary: 700 registered; 701–705 Section 8 accepted; 710 cancelled for Section 8 failure; 800 renewed; 900 expired.

fifth with the lowest, with the fifths cut inside a single Nice class and year so that no difference between industries or between years enters; regression tables add drafting and filer controls and cluster errors on the owner. Where a continuous version is given, it is the change in the outcome for a one-standard-deviation change in the axis, in percentage points.

Counting differs by table. A filing may claim several Nice classes at once, so a *class-record* counts it once per class while a *registration* counts it once by serial number; a *debut filing* is an owner’s first, one per owner. A *cohort* is the set of registrations issued in a calendar year; where a split is instead by filing year, the text says so.

Two windows bound the samples, each set by an institutional fact. Scoring runs 1995–2019: full five-year windows exist on both sides for filings made 1990 through 2020, and scoring holds back five years at the start, where references are thinnest, and one at the end. Five-year-proof cohorts run 2002–2018: cancellations post at about registration age six and a half, the record ends in April 2026, so 2018 is the last cohort whose window has essentially elapsed, and 2002 is the conventional lower cut for this corpus — a bound that matters, because carried back to the first scoreable cohorts the penalty was larger than it has ever been since (Section 6.1).

4 Filing, drafting, and reaching registration

This section is about the first step only: whether an application reaches registration, on every class-record filed 1995–2018. The companion paper treats this step as a validation of the measure (Silver, 2026); what matters here is the contrast it sets up with the steps that follow.

4.1 Registration rises with atypicality and barely moves with lead

Figure 1 gives the registration rate by decile of each axis, with deciles cut within Nice class and filing year so that no difference between industries or between years enters. On atypicality the rate rises from 56.0% in the least atypical decile to 59.1% in the ninth and turns down to 58.3% in the most atypical — a three-point climb and a one-point turn at the top. On lead the range is 0.9 points: filings written in language their class was moving toward register at nearly the same rate as filings written in language it was leaving.

The flatness of lead at this step has a mechanical reading worth stating: lead is computed from the five years of filings *after* the filing date, which had not happened when the application was examined. No participant in prosecution could observe the quantity. Atypicality’s past-facing half, by contrast, is exactly what an examiner steeped in the class’s usual language experiences, and it is the axis registration responds to. The shape — rising, then turning at the extreme — is the one audience-conformity research predicts for evaluators (Hsu, 2006; Zhao et al., 2017; Zuckerman, 1999), with the caution that registration is mostly a measure of the applicant’s own follow-through: of the applications that do not register, roughly nine in ten die by inaction, and only 4.3% are removed adversarially (Table 1).

4.2 Drafting and refiling

Who drafts the description matters for everything that follows. Represented applications are 73% of filings over 1995–2018 and reach registration at 61.9%, against 45.6% for self-filed ones; self-filers write shorter descriptions, sit closer to their class’s usual language, and lean slightly further ahead of it. Because who drafts moves both the language and the outcome at once, counsel of record is held in every five-year-proof regression, and Section 5 reports the estimates separately by drafter.

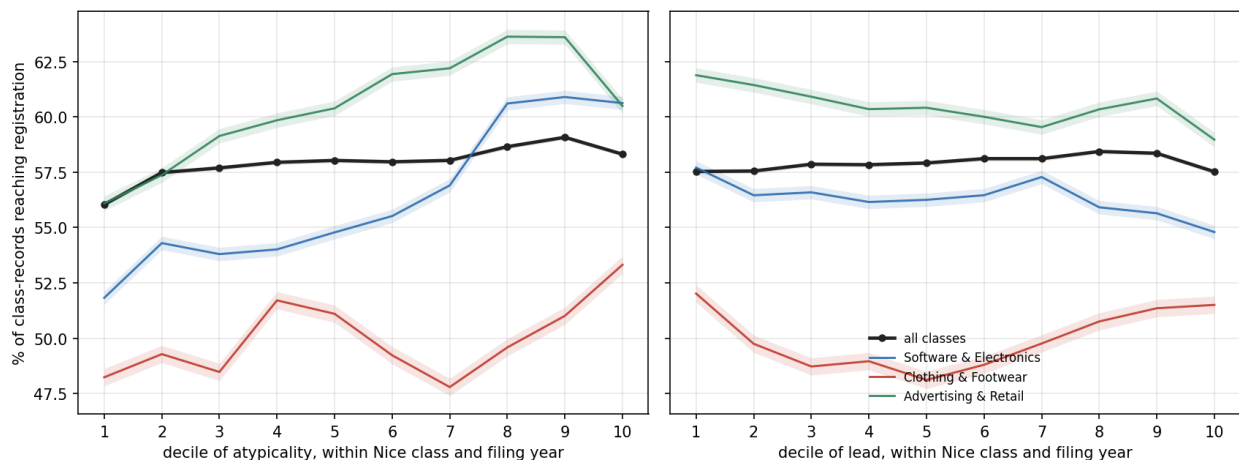


Figure 1: Share of class-records reaching registration, filings 1995–2018, by decile of atypicality (left) and of lead (right), deciles cut within Nice class and filing year; $n = 9,024,532$ scored class-records. The black line pools all 45 classes; the colored lines are three named classes. The grey band is a 95% binomial interval on the pooled line.

Applicants who try again learn to conform. Among 2.28 million abandoned applications, 3.6% are followed within six years by the same owner filing the same mark in the same class, and 1.5% by one that then registers. Of the 42,115 abandoned applications whose owner later registered the same mark in the same class, 74% carry a rewritten description, and the rewrite moves toward the class’s typical language from both tails: originals in the most atypical fifth lose 0.39 nats of atypicality, and originals in the most conventional fifth gain 0.16, whoever drafts the second attempt. Refilings that reuse the identical text move by less than 0.01, so the convergence is in the writing (the companion paper gives the detail by drafter).

Three facts carry forward. Registration is a follow-through indicator, not an examiner’s verdict: nine in ten failures are the applicant’s own inaction. Registration peaks at a middling distance from the class’s usual language and barely moves with lead, which no one at this stage could observe. And applicants who try again conform — while an applicant who conformed before ever filing is unobserved, so that part of the compression cannot be bounded. Every estimate on registered marks in the sections that follow is an estimate on this partly conformed language.

5 Persistence in commerce: the five-year proof of continued use

This section is about one outcome: whether a registered mark’s owner proved, between the fifth and sixth year of the registration, that the product the mark names was still in commerce. The sample is every unique registration issued 2002–2018 that carries a score; failure is a dated cancellation for non-use at registration age 4.0–8.5 years (Section 3.3). Half of registered marks fail it. How the penalty found here moved across industries and decades is Section 6; what became of the firms is Section 7.

5.1 Why the analysis conditions on registration

The question is about companies whose products entered commerce: given that a product was brought to market, does the language of its first description predict whether it was still there five

years on? Registration is the record’s marker of entry — a use-based application asserts the mark is in commerce, and an intent-to-use application registers only after a sworn statement of use — so the analysis begins there. An application that never registered left the record, nine times in ten, by the applicant’s own inaction, before any product met a customer. Including those applications would change the question into one about everyone who considered launching, including those who then did not, and since they are 2.3 million of the 6.0 million scored applications, their inclusion also buries the commercial signal in launch noise. Section 9 shows the pooled analysis for a reader skeptical of the conditioning: the registration step’s shape is stamped onto every later outcome, and the gradients found below are attenuated and in places reversed.

Conditioning would miscount if abandonment were usually a pause rather than an ending — if owners reworked refused applications and refiled, the same product would be counted once as a failure and again as a success. It is not: among 2.28 million abandoned applications, 3.6% are refiled within six years and 1.5% lead to a registration (Section 4.2). Crediting every successful refile as an eventual registration moves the registration profile by less than a tenth of a point. The refiled cases are 1.1% of this section’s sample, they fail the five-year proof at nearly the sample rate (51.7% against 52.8%), and the headline contrast below is +2.25pp with them and without them.

5.2 Leading marks fail the five-year proof more often

Marks written in language their industry subsequently moved toward fail the five-year proof more often than marks written in language it was moving away from. Within Nice class and registration year, controlling for whether the filer drafted the mark themselves or hired counsel, and for the length of the description, registrations in the most leading fifth of lead fail 1.4 points more often (53.3%) than those in the most lagging fifth (51.9%; $t = 8.3$, Table 2).

Among 4.12 million registration class-records 2002–2018, dated cancellation at the five-year proof rises from 50.0% at the bottom lead quintile to 52.4% at the top, positive in 33 of 44 classes with sufficient volume (Figure 2). But a raw pooled contrast mixes the within-industry gradient with class and registration-year composition, treats multi-class registrations as independent observations, and ignores that outcomes correlate within owner. The primary estimates therefore come from a linear probability model on 3.10 million unique registrations, one row per serial, with lead quintiles cut *within* Nice class and registration year. The specification adds class×registration-year fixed effects; controls for counsel of record, intent-to-use basis, log description length, log owner filing volume, owner domicile, and foreign filing basis;² and standard errors clustered on 1.27 million normalized owners (Table 2).³

The gradient is monotone within cells. Whether the filing was drafted by counsel or by the owner is held because it moves both sides of the comparison at once: self-filers fail the five-year proof at 69.6% against 46.8% for represented owners, and Section 4.2 showed they write shorter, more class-typical, slightly more leading descriptions, so without the control part of any lead gradient would be a self-filing gradient. With it, the penalty is of the same order in both strata (+1.8pp self-filed, +1.2pp counselled). The full control set cuts the magnitude by about forty percent without absorbing it — from +2.3pp with fixed effects alone to +1.4pp — and the estimate is present in

²A US application is filed either on *use* — the mark is already in commerce — or on *intent to use*, which defers proof until after allowance. Foreign applicants may instead file on a home registration (§44(e)) or extend an international registration into the US under the Madrid Protocol (§66(a)); the latter maintain under §71 rather than §8. These are statutory routes to the same registration, and they differ sharply in who files and how often the mark is abandoned, which is why each enters as a control.

³Tied scores are common (17.1% of registrations share a lead value within their cell); tie-breaking is deterministic and immaterial, moving the contrast by at most 0.002pp against 2.287 (Supplementary Material §S1).

Table 2: Failure at the five-year proof of continued use and lead: design-based estimates. LPM of event-dated failure on within-class $\times$ registration-year lead quintiles (Q1 omitted), unique serials, registrations 2002–2018. Failure is a dated cancellation for non-use at registration age 4.0–8.5 years, under §8 or, for Madrid §66(a) registrations, §71. All specifications include class $\times$ registration-year fixed effects; controls are counsel of record, intent-to-use basis, log goods/services length, log owner filing count, owner domicile (US, CN), and foreign basis (§44(e), §66(a)). Base failure rate 52.3% on this sample. The same estimates appear in the companion paper, where they validate the measure (Silver, 2026).

Sample	Specification	Q5–Q1 (pp)	SE	n
All	FE only	+2.29	0.16	3,104,534
All	FE + controls	+1.39	0.17	3,104,534
Counsel-represented	FE + controls	+1.15	0.20	2,414,318
Counsel + US-domiciled	FE + controls	+1.60	0.22	1,888,352
Self-filed	FE + controls	+1.80	0.26	690,216
Excluding Madrid §66(a)	FE + controls	+1.82	0.18	2,924,427
Counsel + US, 2002–07	FE + controls	−0.25	0.33	551,989
Counsel + US, 2008–12	FE + controls	+2.00	0.42	551,924
Counsel + US, 2013–18	FE + controls	+2.65	0.25	784,439
All	continuous, pp per sd of lead	+0.58	0.05	3,104,534
Counsel + US	continuous, pp per sd of lead	+0.63	0.06	1,888,352

Standard errors replace significance stars: with n in the millions every coefficient has $p < 0.01$, so stars would mark each row identically. Errors are clustered on *normalized owner* — names uppercased, stripped of punctuation and of two dozen legal suffixes, so ACME CORP. and ACME CORPORATION are one cluster. A single owner files many marks, drafts them in a house style, and abandons them together when the business fails; the within-owner correlation of outcomes at the five-year proof is 0.38 pooled across classes (0.42 in the single-class checks where it was first measured), so the 1.27 million owner clusters, not the filings, carry the information. The Madrid row drops §66(a) registrations rather than absorbing them in a control, since they maintain under a different statute; the estimate is larger without them.

the cleanest stratum, counsel-represented US-domiciled owners (+1.6pp), so it is not carried by the self-filed or foreign mass-filing segments whose failure rates are high for their own reasons.⁴ The penalty is also not constant over the sample: within counsel-represented US-domiciled owners the earliest third of cohorts carries none of it (−0.25pp, SE 0.33) — leading marks registered 2002–2007 were no less likely to survive than lagging ones — while the later thirds carry +2.00 and +2.65pp. Section 6 takes that variation as its subject.

Per-class estimates, with each class’s own profile, are in the online appendix.⁵

The penalty is a rate, not a timing, and not disengagement. The proof is close to a fixed appointment (97% of in-window cancellations post at registration age 6.5–7.0), so a mark failing outside the 4.0–8.5 window would count as a survivor; on the 3.15 million registrations whose ninth anniversary is inside the record, where failure is settled and no window is needed, the contrast is +2.71pp against the window’s +2.65, and conditional on failing, failing late carries a contrast of −0.01pp ($t = -0.2$). Nor is the proof measuring an owner who lost interest: passage is flat across office-action response-speed quintiles (46.1% to 48.1%, medians seven days to 183) and the penalty is present in both speed halves (−3.48 and −2.36pp on passing, against −2.94 pooled).

5.3 The penalty under three theme scorings

Every estimate above scored filings on fifty themes fitted once for the whole corpus, with each class supplying its own reference. The companion paper rescores the identical registrations under *500 global themes* (a theme closer to a product category than to an industry) and under *50 themes fitted separately inside each class* (nothing shared across classes), and the penalty does not depend on the choice: +2.29pp under the production scoring, +2.61pp at 500 themes and +1.94pp under per-class themes, positive under all three at once in 28 of the 44 classes with enough volume, even though the scorings agree only moderately on which filings are leading (lead correlates at 0.64 and 0.52 between them) (Silver, 2026).

Atypicality behaves differently. Under the two global scorings the most atypical fifth of registrations fails the five-year proof 4.3–4.4 points less often than the least atypical; under per-class themes the gap is 0.7 points. The two global scorings’ atypicality correlates at 0.79; global and per-class atypicality correlate at 0.39. A global model places every class in one set of coordinates, so a filing that imports a theme the class rarely uses and other classes use often registers as atypical for its class. A model fitted on the class alone re-partitions the class’s own vocabulary, and the imported theme becomes a local theme like any other. What the global scoring calls atypical is, in large part, drawing on a theme from elsewhere.

That population can be measured directly, and it separates two kinds of new. Under the 500-theme model, for each theme and class, the first sustained year is the first year by which the theme has been the dominant theme of at least one hundred filings in that class; the corpus-wide first sustained year is the earliest of those across classes.⁶ Each registration in the sample is assigned its dominant theme and classified at its filing date as carrying a theme new to the corpus, a theme new to its class (established elsewhere, arriving here), or an established theme. Of 4.12 million registration class-records, 3.97 million carry an established theme, 151,012 a theme new to their class, and 779 a theme new to the corpus — the last too few to support a claim, because almost every theme a 500-theme model can resolve had been established in some class before 2002.

⁴The level effects on those covariates are large in their own right: counsel −19.5pp and Madrid §66(a) basis +10.3pp.

⁵<https://aporia.institute/tm-vocabulary/online-appendix/>

⁶Sustained years and theme shares are computed on a 25% systematic sample of all filings (every fourth serial), scaled up, over filing years 1986–2024.

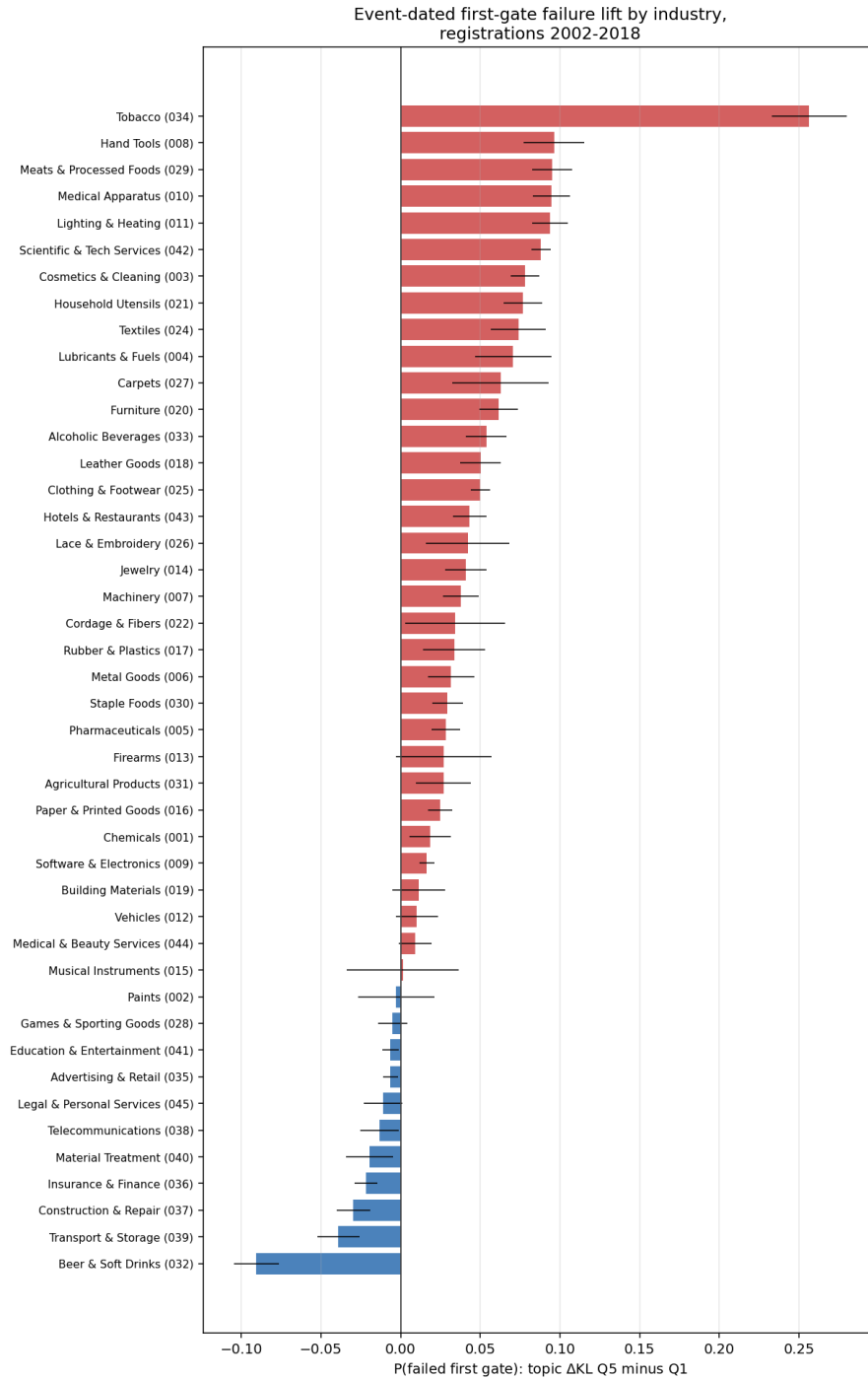


Figure 2: Failure lift at the five-year proof of continued use (lead top minus bottom quintile) by industry, registrations 2002–2018, raw quintile contrasts; classes need 5,000 scored registrations to enter. Positive bars indicate the leading penalty. Whiskers are two-sample binomial intervals, uncorrected for the within-owner correlation (0.38) and so narrower than clustered intervals; read Table 2 for precision. Per-class breakouts are in the online appendix.

Registrations carrying a theme new to their class fail the five-year proof at 48.3% against 51.6% for established themes, and 0.43 points less often within class and registration year (SE 0.13). Holding the filing’s own lead and atypicality as well reverses the sign: net of them, the new-to-class registrations fail 1.06 points more often. The two statements describe one population. A theme arrives in a class in filings that are atypical for that class, and atypical filings survive; that is the -0.43 . Among filings of equal atypicality, the ones carrying the arriving theme fail more; that is the $+1.06$. Together with the collapse of the atypicality gradient under per-class themes, this suggests the protection the global scorings show is largely a reading of themes imported from other classes. The penalty attaches to being early with language, not to being far from the class’s centre.

Nor is the lead penalty an atypicality effect in disguise. The two could be confounded mechanically — lead is the difference of two nonnegative levels whose average is atypicality, so its spread grows with atypicality and its magnitude correlates with it at $+0.29$ — but the signed value does not (-0.05), and cutting lead quintiles within fifths of atypicality leaves the penalty intact in every fifth: $+2.1$, $+3.1$, $+3.3$, $+1.7$ and $+2.5$ pp from the least atypical fifth to the most (each SE under 0.3).

5.4 The risk persists at the year-ten renewal

Survivors of the five-year proof face a combined declaration and renewal at year ten, and on the harmonized estimates of Table 7 the lead penalty persists there at about half its size: -0.97 pp per standard deviation on passing the five-year proof, -0.43 pp on renewing at ten, both stated on survival. Pooled across classes the renewal profile is a V rather than a slope — renewal is highest just below the centre of the lead axis and lowest at both tails, the lagging tail lowest of all (45.5% against 50.6% at the most leading). Most of the difference between leading and lagging resolves at the five-year proof; what remains at renewal is a penalty on both extremes, the lagging one the larger.

5.5 What the five-year proof shows

Among products that entered commerce, being described in language the industry was moving toward predicts being gone within six years: $+1.4$ points on a 52.3% base with the full design, $+2.3$ raw, in three of every four classes, under every partition of the vocabulary, robust to timing, to drafting, and to who files. Atypical language predicts the opposite, but that protection largely disappears when themes are fitted within each class, and filings carrying a theme imported from another class fail more than equally atypical filings. The two steps the record dates respond to the text differently: registration responds to distance from the class’s usual language at a moment when lead does not yet exist to be read (Section 4.1); the five-year proof is recorded after the industry’s language has moved, and it is where being early shows up as a cost.

6 New industries, fads, and waves

This section asks when and where the leading penalty of Section 5 lived: across registration cohorts, across the technology classes, across the themes each era’s leading filings were drawn from, in the filings that name the internet, and inside surging themes. The outcome throughout is the same one — failure at the five-year proof of continued use — and every sample is settled cohorts (registrations whose ninth anniversary is inside the record), so nothing here is censored.

6.1 The penalty weakened in the middle of the sample and was largest at the start

Scores exist from 1995, which makes registration cohorts from 1996 evaluable, and their outcomes at the five-year proof have been settled for two decades. Figure 3 draws, for each cohort, the change in survival for a one-standard-deviation increase in lead and in atypicality, the two entered jointly and standardized within Nice class and cohort. Pooled across classes, the lead coefficient is significantly negative in every settled cohort except 2002–2005 — -1.9 to -2.3 pp per standard deviation for the 1996–2000 cohorts, $+0.4$ to -0.2 across 2002–2005, and -0.5 to -1.4 from 2006 on. In the software class the middle of the sample is not a pause but a reversal: the coefficient runs from -6.8 pp for the 1996 cohort to significantly *positive* ($+0.9$ to $+1.2$) in 2002, 2003 and 2005, then back through -4.0 in 2012. The atypicality coefficient never changes sign anywhere: more atypical registrations survive more in every cohort of both panels, by $+0.6$ to $+2.5$ pp per standard deviation pooled and $+0.6$ to $+5.7$ in software.

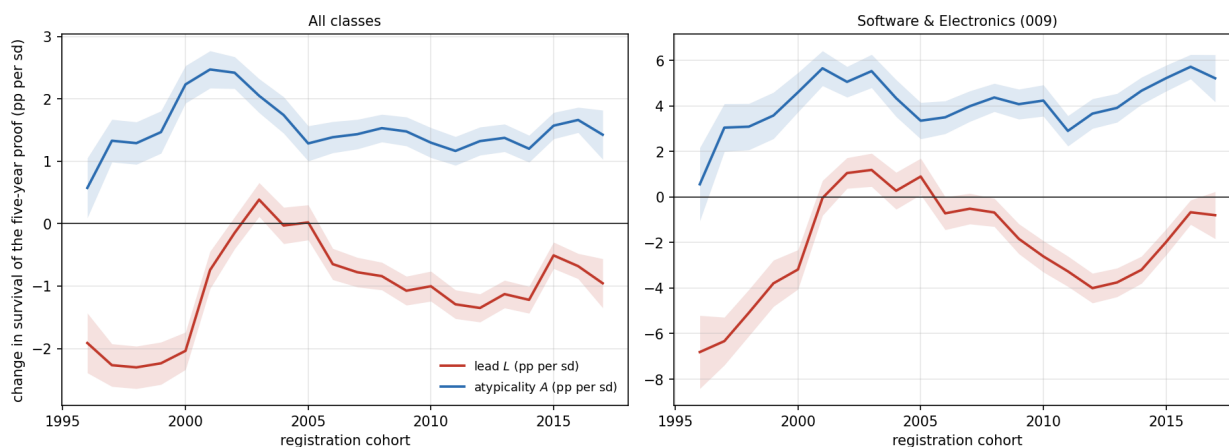


Figure 3: Survival of the five-year proof of continued use, per registration cohort, as continuous coefficients: the change in survival, in percentage points, for a one-standard-deviation increase in lead (red) and in atypicality (blue), entered jointly, both standardized within Nice class and cohort. Left: all classes pooled ($n = 3,150,442$ settled registrations); right: software and electronics alone. Bands are 95% intervals on OLS standard errors, uncorrected for the within-owner correlation of outcomes (0.38), so they understate the uncertainty somewhat.

Stated as cohort contrasts, the swing runs from $+6.9$ pp for the 1996 cohort through $+10.1$ pp for 1999 — roughly twice the largest cohort contrast observed since — to the collapse, and back to $+1.5$ to $+4.7$ pp for every settled cohort from 2008. Neither prosecution-speed selection in the earliest cohorts nor the cohort indexing drives it: holding pendency in a common band leaves the series unchanged, and indexed by filing year the contrast is $+7.9$ to $+9.6$ pp for 1995–1999 filings against -0.95 for filings made in 2000.

Splitting the four classes that carry most technology filings — 009, 035, 038 and 042 — against the other forty-one separates a large moving component from a small steady one (Figure 4):

Filing years	Technology classes	All others
1995–1999	+14.18 (0.36)	+4.48 (0.27)
2000–2004	−1.37 (0.35)	+1.21 (0.24)
2005–2007	+2.04 (0.43)	+1.99 (0.28)
2008–2014	+9.02 (0.25)	+1.32 (0.17)

Outside technology the penalty is positive in every era, largest for late-1990s filings (+4.5pp) and smallest for 2000–2004 filings (+1.2), including the years in which the pooled series reads as zero. Inside technology it swings by fifteen points: strongly positive for filings made into the late 1990s, negative for those made in the five years after, and positive again from 2008. The pooled quiet of 2002–2005 is a technology reversal partly offsetting a non-technology positive, not an absence. (This table cuts quintiles within era and group on the registration-cohort series of Figure 4; Table 3’s raw row, computed on the settled registrations used for the class and theme breakdowns, gives +14.5, −1.8 and +9.2 for the same technology eras.)

What this design cannot say is which cycle the swing belongs to. Marks filed 1995–1999 reach their five-year proof in 2001–2007 and marks filed 2000–2004 reach theirs in 2006–2013, so a story in which unusual language is punished because it was written during an investment boom and a story in which it is punished because the proof falls due during a bust both fit these numbers.

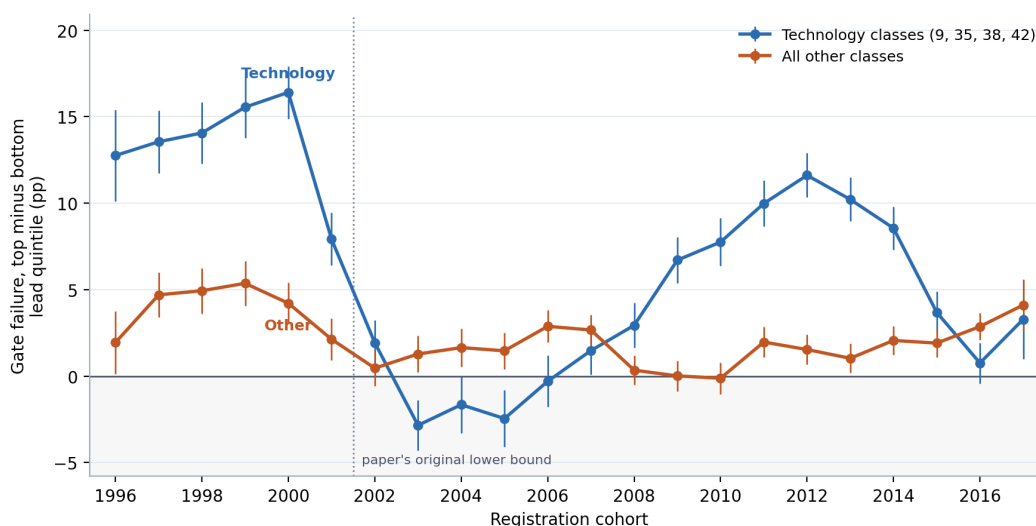


Figure 4: The leading penalty at the five-year proof by registration cohort, technology classes (009, 035, 038, 042) against all others. Each point is the failure rate of the most leading fifth of that cohort’s registrations minus that of the most lagging fifth, lead quintiles cut within cohort and group, no controls, with 95% intervals, on settled registrations. Outside technology the penalty is small and almost flat across two decades; inside technology it swings from +16pp for the 2000 cohort to −3pp for 2003 and back to +12pp for 2012. The dotted line marks the 2002 cut that the cohort samples of Section 5 use.

6.2 Three of the four classes reverse, and the leading language changes theme

The four-class split pools industries that do not move together. Taking them one at a time, and then assigning quintiles inside progressively finer cells (Table 3):

Electrical and software goods (009) and telecommunications (038) follow the pooled pattern. Business services (035), the class that holds online retail, reverses hardest: −9.7pp for filings made 2000–2004 against +4.6pp in the five years before. Scientific and computer services (042) never reverses. Its contrast is +6.7pp in the era in which the other three are negative and between +8 and +17pp in every other, and by registration cohort it stays positive through 2015, turning negative only in the last two settled cohorts (Figure 5). The reversal is an event in goods and in business services; the class that sells technology as a service sat it out.

Table 3: The leading penalty at the five-year proof inside technology, by class and by cell. Top-minus-bottom lead-quintile contrast in failure (pp, SE in parentheses), settled registrations, by filing era. The upper block assigns quintiles within each class and era. The lower block pools the four classes and assigns quintiles within the era (raw), within class $\times$ registration-cohort cells, and within theme $\times$ class $\times$ registration-year cells, where a filing’s theme is the one its description loads most heavily on under the production $T = 50$ model. $n = 1,248,591$.

Class	1995-1999	2000-2004	2005-2007	2008-2014
009 Electrical, software	+18.04 (0.52)	−5.32 (0.51)	+0.25 (0.62)	+11.67 (0.37)
035 Business services	+4.59 (0.69)	−9.70 (0.56)	+2.02 (0.63)	+1.66 (0.36)
038 Telecommunications	+9.91 (1.40)	−4.24 (1.29)	+2.60 (1.48)	+3.42 (0.94)
042 Scientific, computer services	+17.11 (0.57)	+6.70 (0.62)	+8.42 (0.87)	+12.63 (0.47)
Four classes pooled, raw	+14.49 (0.33)	−1.81 (0.31)	+2.56 (0.38)	+9.16 (0.22)
within class $\times$ cohort	+14.27 (0.33)	−3.63 (0.31)	+2.73 (0.38)	+7.24 (0.22)
within theme $\times$ class $\times$ cohort	+4.78 (0.33)	−1.26 (0.31)	+2.56 (0.38)	+4.51 (0.22)

Assigning quintiles within class and registration year rather than within the era pool changes little. Assigning them within theme, class and registration year changes a great deal: the late-1990s contrast falls from +14.5 to +4.8pp, the post-2008 contrast from +9.2 to +4.5pp, and the 2000–2004 reversal from −3.6 to −1.3pp. Two thirds of the first and half of the second are between themes rather than within them. The leading fifth of each era was drawn from particular themes, and those themes carried their own failure rates. The penalty for being leading *inside* a theme is steadier: +2.6 to +4.8pp in every era except 2000–2004, when it was −1.3, the same order as the penalty outside technology.

Which themes is a matter of record (Table 4; the table’s last row is the all-themes pooled contrast within theme $\times$ class $\times$ cohort). In 1995–1999 the computer, software, information and data services theme held 38% of technology filings and 75% of the leading fifth, and failed at 64.9% — that one theme is most of the late-1990s penalty. In 2000–2004 the leading language had moved to electronic media while the computer and information-services wording fell out of the leading fifth, and inside several themes the lagging fifth now failed more than the leading one. From 2008 the leading fifth over-represents downloadable and online software four to one — the application store’s vocabulary, the highest-failure theme (67.2%), with a within-theme contrast of +3.3pp in 2008–2014. The full theme tour is in Supplementary Material §S2.

So the swing has two layers. In most eras, a leading filing is a few points more likely to fail than a lagging one in the same theme, class and cohort, in technology as elsewhere. On top of that, each era’s leading vocabulary was concentrated in one or two themes whose marks failed at their own rate: computer and information services in the 1990s, application software after 2008. In the trough between, the language that had been leading fell out of the leading fifth, and the marks still written in it failed at high rates (Section 6.4). The design-based estimates of Table 2 hold class and cohort fixed but not theme, so they carry both layers; a reader who wants only the within-theme penalty should take the lower block of Table 3 as the guide to its size.

6.3 Internet as a theme: web-based filings against their class

A filing is internet-bearing when its goods/services text names the internet, the web, a website, online or on-line provision, or electronic commerce. The rule reads the text rather than the class, so a web retailer filing in apparel is caught and a semiconductor maker filing in class 009 is not.

Table 4: The leading penalty at the five-year proof by theme within the four technology classes. Themes with at least 20,000 settled registrations, labelled by their five highest-probability terms; quintiles assigned within theme and era. Cells with fewer than 3,000 registrations are blank. The final row pools all themes, with quintiles cut within theme×class×cohort.

Theme (top words)	<i>n</i>	1995-1999	2000-2004	2005-2007	2008-2014
services, computer, software, information, data	451,253	+3.61 (0.52)	+3.56 (0.49)	+4.18 (0.64)	+5.92 (0.37)
video, computer, electronic, music, games	162,290	+2.53 (0.97)	−3.62 (0.89)	−2.85 (1.01)	+3.36 (0.60)
services, featuring, store, retail, store services	131,107	+8.39 (1.09)	−5.49 (1.01)	+2.92 (1.15)	+0.98 (0.66)
services, field, training, estate, educational	95,602	+7.82 (1.09)	+0.99 (1.12)	+6.02 (1.43)	+2.05 (0.83)
apparatus, systems, devices, electronic, control	84,121	+4.80 (1.27)	−6.92 (1.19)	−0.58 (1.41)	+3.65 (0.88)
design, field, systems, development, services	32,517	+10.40 (2.03)	+4.67 (1.82)	+1.50 (2.27)	−2.27 (1.45)
software, online, downloadable, non-downloadable, virtual	29,472	—	—	—	+3.30 (1.05)
bags, clothing, shirts, jackets, leather	28,421	+1.28 (2.59)	−1.55 (2.39)	−4.66 (2.40)	+6.06 (1.36)
research, medical, scientific, testing, purposes	25,897	+12.65 (2.66)	+12.09 (2.03)	−3.82 (2.59)	+5.63 (1.51)
services, health, field, care, information	22,880	+4.94 (2.04)	+1.39 (2.36)	—	+3.50 (1.72)
All themes, within theme×class×cohort	1,150,921	+4.78 (0.33)	−1.26 (0.31)	+2.56 (0.38)	+4.51 (0.22)

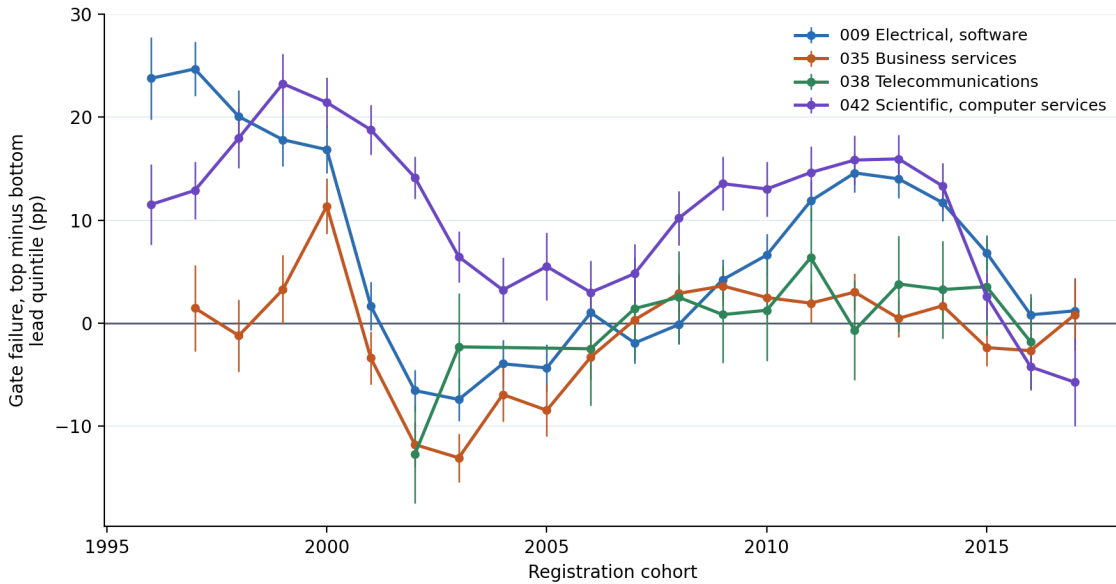


Figure 5: The leading penalty at the five-year proof by registration cohort, one line per technology class. Each point is the failure rate of the most leading fifth of that class and cohort’s registrations minus that of the most lagging fifth — lead quintiles cut within class and cohort, no controls — with 95% intervals, on settled registrations. Three classes cross zero between 2001 and 2006; scientific and computer services does not.

Internet-bearing filings are 16.9% of scored registrations: 58% in telecommunications, 39–41% in advertising and retail, scientific and technical services, and legal and personal services, under 5% in most goods classes. Figure 6 puts every class on two axes at once — how much more often its internet-bearing filings register, and how much more often they fail the five-year proof; Supplementary Material §S2 tabulates the underlying rates for every class.

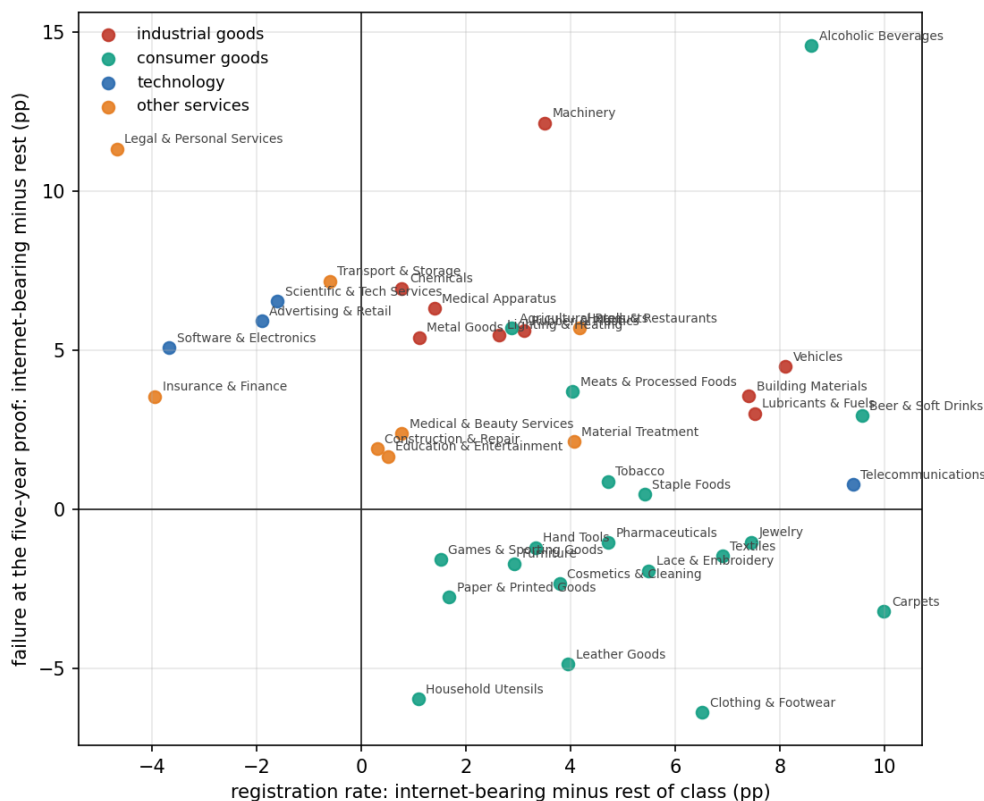


Figure 6: Each Nice class’s internet-bearing filings against the rest of the class, on both steps at once. Horizontal: registration rate of internet-bearing class-records minus the rest of the class, filings 1995–2018. Vertical: failure at the five-year proof of continued use, internet-bearing minus the rest, registrations 2002–2018. Classes with fewer than 500 internet-bearing registrations are dropped. Classes in the upper-right quadrant grant internet filings more often and lose them more often.

Three patterns are in the figure. In the industrial goods classes, internet-bearing registrations fail the five-year proof far more often than their class: machinery 50.9% against 38.7%, chemicals 46.1% against 39.1%, vehicles 49.6% against 45.1%. A web-based business registering in an engineering class fails at the rate of a software registration (51.4%), not at the rate of its class. In the consumer goods classes the sign reverses — internet-bearing registrations in clothing fail 52.4% against 58.7% for the class, in household utensils 44.5% against 50.5% — so online sellers of consumer goods outlast the class’s other registrants. In every service class the internet-bearing filings fail more often, by up to eleven points, most in legal and personal services (60.1% against 48.7%). At the registration step the goods classes sit to the right of zero: pooled across them, internet-bearing applications reach registration 59.5% of the time against 56.0% for the rest. For the industrial-goods classes the two steps therefore move in opposite directions — more likely to be granted, then more likely to be cancelled for non-use six years later — the pattern Section 5.5

describes, in which registration responds to distance from the class’s language and the proof to whether the product stayed in commerce.

The failure gap has a common shape in event time. Dating each class’s internet arrival at the first filing year the pattern reaches 2% of its filings, the gap starts near +7 points in the first observable cohorts and settles between +1 and +4 nine to eleven years after arrival, shrinking in six of the eight classes observable in both phases; calendar time disguises this, since machinery’s arrival was 2008 and its peak 2012–2015 (Supplementary Material §S2). Each class’s clock appears to start when the vocabulary arrives in it rather than when the internet did, a shape consistent with falling entry costs followed by normalization.

The internet split also locates the era swing of Section 6.1. Rerunning the per-cohort lead coefficients of Figure 3 separately for internet-bearing and other registrations (Figure 7): the mid-sample reversal belongs to the internet-bearing filings, whose leading registrations survived *more* in the 2001–2005 cohorts (+0.8 to +2.5pp per standard deviation) while non-internet technology registrations sat near zero. The other two episodes are not the internet’s in this textual sense. The late-1990s penalty is present in technology registrations that never name the web (−3.4 to −4.8pp for the 1996–2000 cohorts) — consistent with Table 4, where that era’s deadly vocabulary is computer and software services, words that predate explicit internet naming. And the post-2008 penalty sits in both splits at once (−2 to −3pp each). Outside technology, the non-internet series stays between 0 and −1.4pp throughout. What the internet changed, on this cut, is the middle: for half a decade after the crash, the web filings whose language led their class were the ones that lasted.

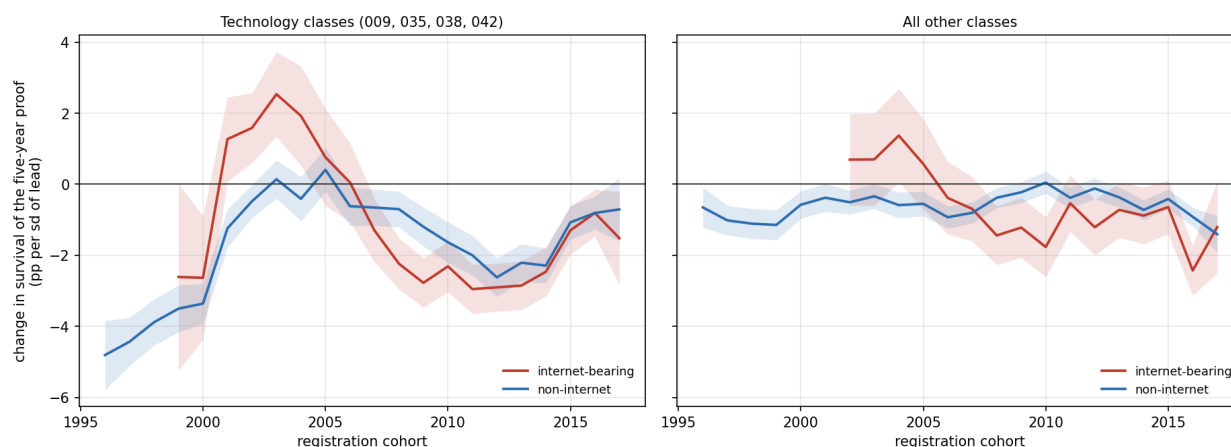


Figure 7: The per-cohort lead coefficient of Figure 3, split by the curated internet text pattern: change in survival of the five-year proof per standard deviation of lead, atypicality held, standardized within Nice class and cohort. Left: the four technology classes; right: all other classes. Bands are 95% intervals on OLS standard errors; cohort cells under 2,000 registrations are dropped, which is why the internet-bearing lines start late.

The Nice Classification has twice faced this question and twice declined to make the internet a class.⁷ The survival pattern bears on the choice: internet-bearing filings in a goods class fail at rates closer to the technology classes than to their own, so the technology overlay carries more information for this outcome than the class assignment — and Nice classes used as industry controls, as this paper and the literature use them, do not partition the internet economy.

⁷The eighth edition (2002) split the old class 42 into classes 42–45 by kind of service; the twelfth (2023) placed downloadable virtual goods in class 9 and distributed metaverse services across the existing service classes.

6.4 What carried the reversal: a fading theme, and rewards in goods and online retail

The mid-sample reversal can be taken apart with the two instruments of this section held at once: each 2000–2004 technology filing carries its dominant theme and its internet flag, and the pooled lead quintiles of Section 6.1 can be re-read cell by cell (Table 5).

The sign flip itself is carried by one theme, and not by reward. By 2000–2004 the computer, software, information and data services theme — the vocabulary that had supplied three quarters of the late-1990s leading fifth (Table 4) — had largely left the leading fifth: it still held 40% of the era’s filings but only 11% of its leading fifth, and its marks failed at 57% and 61%, among the era’s highest rates. Excluding that one theme flips the pooled contrast from -1.8pp back to $+0.9$. At the pool level, then, much of the reversal is the record of marks still written in a fading vocabulary failing away from the leading end. And inside the theme, being leading was never rewarded: $+2.6\text{pp}$ among its other filings, $+1.8$ among its internet-bearing ones, in this era as in every other.

The within-cell reward is real, and it sits in goods and retail themes. Within class and cohort, internet-bearing filings show -4.5pp against -2.9 for the rest. Cell by cell, the strongest single reward is in the internet-bearing half of the retail store services theme — online retail, -8.3pp . The other large rewards are in filings that do not name the internet: the apparatus and devices theme (-7.1), electronic media (-3.2), and the non-internet half of retail store services (-4.2). Where filings described research and scientific services, the penalty never left: that theme penalized its leading fifth by $+11.8\text{pp}$ in the same years.

What the cells show is where the reward lived; they do not show why. A reading consistent with them is that in the years after the crash, leading language paid where it described goods and the channels that sold them — devices, media, and online retail most of all — while the self-described information services that had led the boom were the crowded language of a wave that had broken. The design cannot separate that reading from its selection twin, that the firms still filing in 2001–2003 were already unusual survivors of the shakeout; and the boom-and-bust ambiguity of Section 6.1 applies unchanged, since these filings reached their proofs in 2006–2013.

6.5 Surges: joining a wave

Genuinely new themes are rare (Section 5.3); the observable event is a surge, an established theme’s share doubling, over its own three prior years, to a level that matters. The one corpus-scale episode — online software doubling to 2.4% of all filings in 1999 — failed the five-year proof 8.6 points above class and year, 8.5 net of the filings’ own lead and atypicality, with no gradient in arrival order inside the wave (-0.2pp , SE 1.9). Class-scale surges are common (962 episodes at a 1% within-class threshold) and cost little pooled: $+0.65\text{pp}$ within class and year, $+1.27$ net of lead and atypicality, with episodes differing in sign among themselves. Within waves, entry order does not matter; larger waves are not deadlier; and growing waves carry more excess failure than receding ones ($+4.3$ against $+1.0$ points, suggestive on 61 episodes) — a pattern consistent with selection at the proof rather than crowding out. The surge figure, the episode table, and the wave-level facts are in Supplementary Material §S2.

6.6 What the waves show

The leading penalty is positive in every era outside technology and episodic inside it, and much of the episode is between themes: each era’s leading vocabulary was concentrated in one or two themes that failed at their own rate, while the within-theme penalty holds at $+2.6$ to $+4.8\text{pp}$ in every era except 2000–2004. Web-based filings fail hardest in each class’s own first internet years

Table 5: The 2000–2004 reversal, cell by cell. Technology-class registrations filed 2000–2004 and settled at the five-year proof; each carries its dominant $T = 50$ theme and the internet text flag of Section 6.3. Every theme $\times$ flag cell holding at least 5,000 registrations is shown. *Share* is the cell’s share of the era’s filings; *of top fifth*, its share of the era’s pooled leading fifth. Q5–Q1 is the failure contrast between the most leading and most lagging fifths, quintiles reassigned within the cell; in the bottom block, quintiles are pooled across the era (first two rows) or cut within class $\times$ cohort (last two).

Cell	n	Share	Of top fifth	Failure	Q5–Q1
services, computer, software, information (other)	62,143	24.5%	6.3%	57.3%	+2.61 (0.63)
services, computer, software, information (internet)	38,235	15.1%	5.2%	61.4%	+1.78 (0.79)
video, computer, electronic, music (other)	25,042	9.9%	27.8%	56.1%	−3.24 (0.99)
apparatus, systems, devices, electronic (other)	17,032	6.7%	2.6%	45.6%	−7.06 (1.21)
services, field, training, estate (other)	16,349	6.4%	8.9%	44.3%	+1.33 (1.23)
services, featuring, store, retail (other)	15,760	6.2%	1.5%	52.6%	−4.19 (1.25)
services, featuring, store, retail (internet)	8,365	3.3%	0.6%	54.9%	−8.25 (1.72)
design, field, systems, development (other)	7,010	2.8%	7.4%	51.8%	+4.42 (1.89)
video, computer, electronic, music (internet)	5,758	2.3%	3.3%	55.5%	−2.04 (2.08)
research, medical, scientific, testing (other)	5,560	2.2%	5.3%	51.9%	+11.78 (2.10)
All filings, pooled quintiles	253,714	100%	100%	53.4%	−1.81 (0.31)
excluding the services theme	153,336				+0.90 (0.40)
Internet-bearing, within class $\times$ cohort	63,190			58.1%	−4.52 (0.62)
Other filings, within class $\times$ cohort	190,524			51.8%	−2.87 (0.36)

and converge toward their class within a decade. Joining a surging theme costs little on average and nothing for arriving early within the wave; the one corpus-scale wave, online software in 1999, is the exception at every margin. What a fad costs, on this record, is mostly a matter of which theme the fad was, not of when the filer joined it.

7 Commercial success: funding, listing, and the firms that mattered

A mark that survived the five-year proof names a product that held its place in commerce. This section asks whether the owners of leading or atypical debut filings went on to the outcomes that make a firm rich: a priced private round, SEC financial reporting, a public listing, and revenue at scale. The unit is the owner’s first registered filing, its debut, scored on the production measure; the outcomes are dated events in the SEC record. Form D became mandatory in electronic form on 16 March 2009, so the funding ladder is built on debuts made 2009–2018, for which every rung is observed; the public-reporting analysis uses the longer 1995–2018 sample.

7.1 The ladder: a round, reporting, a listing

Of 999,442 owners whose first registered filing, carrying a score, was made 2009–2018, 15,991 (1.60%) raised a priced private round after the debut, 4,154 (0.42%) appear in SEC reporting records at any time, and 1,669 (0.17%) registered a class of securities or completed an initial public offering at any time. The funded rung requires the round to postdate the debut; the reporting and listing rungs do not, so owners already reporting before their first trademark filing are counted, and the staged analysis in Section 8 shows they are a majority of the exchange-listed. The rungs nest: 6.9% of funded owners reach reporting against 0.3% of the unfunded, and 40% of reporting owners reach an IPO.

Table 6 gives the rate at each rung by fifth of lead and of atypicality, with fifths cut within Nice class, debut year, and fifth of description length, so that neither the industry, the year, nor the length of the description enters the contrast. Funding does not vary with either axis: the most leading fifth of debuts is funded at 1.69% against 1.74% for the most lagging, and the most atypical at 1.51% against 1.57%, both within two standard errors. Trademark counts and breadth carry a funding signal (Block et al., 2014); vocabulary position does not. Reporting and listing do vary. Owners whose debut language was most lagging reach reporting at 0.55%, owners whose debut language was most leading at 0.36%; for an IPO the rates are 0.22% and 0.17%. On atypicality the sign is positive on this sample: the most atypical fifth reports at 0.51% against 0.39% for the most conventional, and lists at 0.20% against 0.14%. Section 7.3 absorbs description length continuously on 1995–2018 debuts and finds the atypicality gradient flat; with length held in fifths on 2009–2018 debuts it is positive at t above 4. The two are stated side by side; the difference between them is a length control and a sample, and the atypicality result at the listing margin is the one claim in the paper that depends on which is used.

The lead profile at reporting and at IPO is not monotone: it falls from the lagging end through the fourth fifth and turns up in the fifth. Owners whose first description used language the class was leaving behind are the likeliest to report and to list. Most of them are established firms making a first trademark filing late in a business already built on the class’s older vocabulary, and the ladder cannot separate that from anything the language itself does. Listings that arrive without an observed post-debut Form D round lean the same way: their owners’ debut lead averages -0.22 standard deviations against -0.07 for funded listings — consistent with established, revenue-

Table 6: The ladder by fifth of debut lead and of debut atypicality. Rates in percent of owners whose first registered filing was made 2009–2018 ($n = 999,442$); fifths cut within Nice class, debut year, and fifth of description length. The last column is the most-minus-least contrast with its t -statistic.

Rung	Axis	Q1	Q2	Q3	Q4	Q5	Q5–Q1 (pp; t)
Funded	lead	1.74	1.61	1.50	1.46	1.69	–0.05 (–1.3)
	atypicality	1.57	1.59	1.68	1.65	1.51	–0.06 (–1.7)
Reporting	lead	0.55	0.45	0.39	0.32	0.36	–0.19 (–9.1)
	atypicality	0.39	0.34	0.41	0.43	0.51	+0.11 (+5.3)
IPO	lead	0.22	0.19	0.13	0.12	0.17	–0.06 (–4.0)
	atypicality	0.14	0.14	0.16	0.19	0.20	+0.06 (+4.9)

financed businesses listing without venture money, where “unfunded” means only that no Form D notice is observed.

7.2 The firms that produced the most value

Among the 1.80 million owners whose first registered filing was made 1995–2018, 4,710 link to SEC financial statements with reported revenue; ten of them hold 18.5% of the \$19.2 trillion of linked peak revenue and five hundred hold 82.9%. The rates above do not carry over to this top tail. The ten largest sit 0.35 standard deviations below average atypicality (not significantly) and at average lead; the five hundred largest sit slightly above average atypicality and at average lead; and leading debuts are, if anything, over-represented among the fifty and hundred largest, a quarter or more of which debuted in the most leading fifth. Weighting the whole linked set by revenue puts it at the average on both axes (tier detail in Supplementary Material §S3). The lead gradient is a gradient in rates of reaching the capital markets; among the few firms that became very large, having been early is neither a mark of success nor an obstacle to it. Hand-curated vocabularies show a different split: debuts in AI or blockchain vocabulary are funded roughly five times as often as the average debut (7.8% and 8.0% against 1.6%), while at the IPO rung the gap disappears (0.16–0.22% against 0.17% overall), on very small counts (Supplementary Material §S3).

7.3 Leading debuts reach public reporting less often; unusual ones no more often

The longer 1995–2018 sample tells the same story at the reporting rung, under the design-based estimates the companion paper reports as a validation of the measure (Silver, 2026). The outcome is whether an owner ever appears in the Securities and Exchange Commission’s financial reporting records — broader than exchange-listed, since roughly half the matched firms carry a ticker and the rest report for other reasons. Raw reporting rates rise steeply with atypicality, but the gradient is composition: within class and filing year, with description length held, the most atypical fifth of debuts reaches SEC reporting at the same rate as the least (+0.000pp on a 0.481% base, SE 0.018). Lead is different. At equal atypicality, being far from the class in either direction goes with reporting (+0.067pp per standard deviation of lead magnitude, $t = 7.7$), but being on the leading side goes against it (–0.047pp per standard deviation of signed lead, $t = -7.8$), and the most lagging fifth out-reports the most leading in four of the five atypicality strata. Splitting matched owners on whether their SEC filer number carries a ticker gives 10,556 listed against 9,333 reporting without one, and both halves show the same pattern in lead, so the result does not

depend on counting non-operating registrants as successes. Owners are matched to SEC registrants by normalized exact name, which bounds the atypicality reading in particular: distinctive names match more easily, and name distinctiveness plausibly travels with text distinctiveness.

7.4 Leading does not pay, even where novelty does

The answer to this section’s question is no. An owner whose surviving product had been described ahead of its industry’s language is no more likely to raise a priced round, and reaches SEC reporting and listing *less* often than a lagging owner. What is associated with reaching the reporting margin is unsigned: lead magnitude and description length, which plausibly mark a specific, carefully drafted first filing, carry positive coefficients while the signed component stays negative. Within the firms that do both, vocabulary position and patenting are only weakly related (every coefficient under 0.06 standard deviations; Silver, 2026), so none of this is patenting seen through a different record. Being unusual is at worst free and at best mildly rewarded; having been early is not rewarded on average at any rung the record shows, and the listings that arrive without observed venture money belong disproportionately to the lagging. Among the handful of firms that grew very large, the sign does not carry over: having been early neither marks them nor held them back.

8 Discussion: each axis bites at a different stage

Table 7 re-estimates every outcome on one harmonized sample — one row per debut owner, scored on that owner’s first filing — under a single specification, so the coefficients can be read down a column. Magnitudes differ from the section-specific tables above.

Table 7: Every outcome in the funnel, conditioned on its prerequisite, under one specification. Unit is the debut owner, scored on that owner’s first filing. Each row is a linear probability model with class×debut-year fixed effects and heteroskedasticity-robust errors; coefficients are percentage points per standard deviation. *Atypicality* is the average of the two KL levels; *lead* is the signed difference; the lead-magnitude column $|L|$ comes from a second specification that adds it. Cohort windows differ because each maintenance deadline needs elapsed time and Form D is electronic only from 2009, so the rows are not a nested decomposition. The five-year proof row is survival of a dated cancellation for non-use; the year-ten row is whether a registration that passed the proof was renewed rather than cancelled or expired, restricted to 2002–2013 cohorts, since a year-six and a year-ten cancellation share a terminal status and are separable only by date.

Outcome (<i>population</i>)	<i>n</i>	Base	Unsigned		Signed
			Atyp.	$ L $	Lead
Reached registration (<i>filed</i>)	3,161,552	56.11%	−0.982 (0.032)	−0.497 (0.047)	+0.475 (0.028)
Passed the five-year proof (<i>registered</i>)	1,260,392	41.82%	+1.311 (0.053)	−0.527 (0.083)	−0.968 (0.051)
Renewed at year ten (<i>passed the proof</i>)	297,944	55.92%	+1.383 (0.104)	−0.221 (0.156)	−0.428 (0.096)
Raised a Reg D round (<i>filed</i>)	1,535,004	2.38%	−0.125 (0.015)	+0.345 (0.027)	−0.393 (0.017)
Raised a Reg D round (<i>registered</i>)	916,448	2.45%	−0.169 (0.020)	+0.329 (0.035)	−0.324 (0.022)
Listed after the round (<i>funded</i>)	36,475	3.51%	+0.866 (0.138)	+0.734 (0.195)	−0.534 (0.129)
Became SEC-reporting (<i>registered</i>)	1,773,938	0.56%	+0.017 (0.007)	+0.023 (0.012)	−0.018 (0.007)

Standard errors in parentheses, in place of significance stars; see the note to Table 2.

Unusual language is penalized at registration and rewarded afterwards. With description length held, as it is throughout this table, atypical debut filings complete registration less often (−0.98pp per standard deviation) — unlike the decile profile of Section 4.1, which does not hold length

and rises — then survive the five-year proof of continued use more often (+1.31pp) and reach SEC reporting slightly more often (+0.017pp on a 0.56% base). Having been early runs the other way: leading applications complete registration slightly *more* often (+0.47pp) and then pass the five-year proof *less* often (−0.97pp on passing). Both survival rows point the same way at the year-ten renewal, four years later and on about a quarter of the sample: atypicality +1.38pp and lead −0.43pp among marks that had already cleared the five-year proof. Whatever the five-year proof selects on, the renewal selects on it again rather than reversing it.

In the financing rows the two axes behave differently. Atypical debut owners raise a Regulation D round slightly *less* often (−0.17pp per standard deviation against a 2.45% base), yet among owners who did raise one, the same measure predicts *reaching* the public market (+0.87pp on a 3.51% base, $t = 6.3$). The comparison is among survivors, so it cannot separate a reading in which investors pass over information that is free and public at the moment of filing from selection on unobservables. Three limits bound these rows: Regulation D notices are electronic only from 2009; owners are matched by normalized name, which succeeds for a minority favouring formally constituted entities; and the listing row conditions on funding realized after the filing being scored, so it establishes that information distinguishing eventual listers was present in the text years earlier, not that the language caused the outcome.

Lead does not reverse anywhere after registration: negative at the five-year proof and the year-ten renewal, at financing with or without registration imposed, at listing among the funded, and at SEC reporting among registered owners. Leading filings fare worse at every stage after the one at which lead could not yet be observed. (In the quintile contrasts of Table 6, funding is flat across fifths of lead. The two summaries differ in form: the ladder compares fifths cut within class, year and length, while this table fits a linear term with length held continuously.)

Why would having been early cost? The two axes are not independent in the obvious economic sense. What makes a filing leading is precisely that its industry moved toward its language, and an industry that has moved toward your language now describes its goods the way you describe yours. Being early therefore consumes the quantity that pays: the mark that was unusual in 2011 is ordinary by 2016 not because its owner changed anything but because the category arrived. On this reading atypicality is a position competitors erode, and lead measures how fast. The design cannot demonstrate that mechanism; what can be said is that a negative coefficient on lead at equal atypicality is what the mechanism would produce if it were real.

A second reading follows from the first, and the design cannot test it either. Privately, writing ahead of the industry’s language is hard to rationalize: no stage of the record rewards it, and each era’s leading vocabulary drew filers by the thousands anyway. Socially, the exploration is not obviously waste. The corpus moves — what makes a filing leading is precisely that its class later wrote the way it did — so the language every industry eventually standardized on was first filed by entrants who disproportionately did not survive to use it. If finding the vocabulary an industry will adopt has value to the eventual adopters, that value accrued to the followers while the cost fell on the explorers.

8.1 Limits

Three limits bound the reading. The vocabulary is positional rather than causal: a leading filing was ahead of its industry’s language, which implies a trajectory without asserting that the filing caused it, and a filing can sit where its class was heading because it was imitated, because both responded to the same outside event, or by coincidence. What drives the episodic pattern of Section 6 is not settled: cohort and proof year move together, so post-crisis composition, app-era speculative filings and calendar-time enforcement changes cannot be separated on this design. And

the registered description is a conformed one: applicants who refile rewrite toward the class from both tails, and an applicant who conformed before ever filing is unobserved, so every estimate on registered marks is an estimate on prosecution-filtered language, and the compression cannot be bounded from below.

9 Robustness: what changes under other choices

Each result above is stated under one formulation chosen for legibility: one theme resolution, flat weighting inside the reference windows, symmetric five-year windows, estimates conditional on grant, and the 2002–2018 registration cohorts. The table summarizes what each choice costs; the paragraphs carry the four checks a reader will probe, and the companion paper’s supplement carries the full treatment of everything else — functional forms, the term-scored replication, the listing-versus-reporting split, the window-mix decomposition, and the anatomy of L as a small residual of two large levels (Silver, 2026). No sign of the headline contrast depends on any of these choices.

Unless a row states its own, the baseline is the five-year-proof lead contrast, +2.29pp (atypicality −4.39pp), registrations 2002–2018; entries are the variant’s value in pp.

Variant	Result	Reading
Terms for themes (flat / decayed)	+3.56 / +2.92	conservative representation
Decayed reference windows	+1.95	the one estimate that moves
Resolution, class 009: $T = 50$	+3.23	sign fixed across the sweep
$T = 100$	+2.71	
$T = 200$	+5.20	
$T = 500$	+5.69	magnitude a band, not a point
Corpus-wide at $T = 200 / 500$	+2.23 / +2.61	pooling steadies it
Second seed, class 009, $T = 50$	+3.89 (from +3.23)	scores noisy, sign stable
$T = 200$	+4.88 (from +5.20)	
Window width 2–7 years, class 009	+2.02 to +3.37	flat from five years
Window mix, foresight / rupture	+1.65 / +3.00	rupture the stronger sense
Proof window 3.5–9.0 / 4.5–8.0	+2.29 / +2.28	window immaterial
Cohorts 2002–09 / 2010–18	+0.98 / +3.11	the era swing of §6.1
Excluding duplicate-text filings	+3.05; atyp. −5.68	boilerplate dilutes
Excluding Manual-scale texts only	+2.64; atyp. −5.05	same
Atypicality, cohort halves	−4.55 / −4.28	stable
Unconditional (no grant filter)	composite 22.4 → 22.1%	conditioning exposes, not creates
Ladder, counsel / self (base −0.19)	−0.26 / −0.03	gradient is counsel-borne
Ladder halves (base atyp. +0.11)	0.00 / +0.19	belongs to the late half

Representation and weighting. Four per-filing builds exist: themes or terms, flat or two-year-half-life weighting. Weighting is close to irrelevant (lead correlates at 0.97–0.98 across it, atypicality at 1.000); representation is the real test (themes and terms agree at only 0.46 on lead), and it is the conservative one — term scoring returns larger lead contrasts (+3.56 and +2.92pp) and smaller atypicality contrasts (−3.63, −3.73) than the production build’s +2.29 and −4.39.

Resolution and fitting noise. Re-fitting the theme model at $T \in \{50, 100, 200, 500\}$ leaves the sign and significance of the software-class contrast untouched while the magnitude moves non-monotonically between +2.7 and +5.7pp; corpus-wide, the three resolutions scored for every class agree at +2.29, +2.23 and +2.61pp. A second fit differing only in its random seed reproduces per-filing lead at $r = 0.72$ at fifty themes and 0.79 at 200, while moving the software-class estimate by well under a point in either direction. Each fit is a noisy measurement of lead, and that noise attenuates every lead coefficient toward zero rather than manufacturing it.

Decayed windows. Weighting recent language up with a two-year half-life is the one choice that shifts the headline: +1.95pp against +2.29, the same sign, precision, and profile. The flat window is kept as the simpler statement; a reader who prefers recency should carry +1.95.

Conditioning and composition. Without the grant filter, the composite outcome (registered and passed the proof) is nearly flat in lead because two opposed gradients cancel — registration rising, survival falling — so conditioning exposes the market’s gradient rather than creating it. Excluding the 47% of registrations whose exact normalized text is shared with another filing *strengthens* both headline gradients (lead +3.05pp, atypicality −5.68); identical texts carry identical scores and cluster at the typical end, diluting both. The non-monotone profiles on signed lead decompose into a symmetric part inherited from atypicality composition, a symmetric conditional lead-magnitude effect, and a small linear tilt whose fitted vertices land where that arithmetic puts them (companion paper, supplement).

Two cautions carry across every raw contrast: outcomes correlate within owner at 0.38, so unclustered *t*-statistics in the per-class, per-era, and per-episode cells overstate precision by roughly a factor of two; and the penalty is not constant over the period, so the pooled row is an average over the swing of Section 6.1, and a reader who wants the penalty for a period should read Figure 3 rather than the pooled estimate.

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Data availability

All code and every generated figure and table are public at <https://github.com/ericssilver/tm-vocabulary>; the derived data tables (dated prosecution events, event-dated outcomes, the theme model, and per-filing scores) are published as the repository’s **data-v1** release, and an interactive per-class online appendix is at <https://aporia.institute/tm-vocabulary/online-appendix/>. The underlying USPTO, SEC, and PatentsView sources are public.

Declaration of generative AI in the writing process

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References

- Barron, A. T., Huang, J., Spang, R. L., & DeDeo, S. (2018). Individuals, institutions, and innovation in the debates of the French Revolution. *Proceedings of the National Academy of Sciences*, 115(18), 4607–4612.
- Berkes, E., & Gaetani, R. (2021). The geography of unconventional innovation. *The Economic Journal*, 131(636), 1466–1514.
- Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent Dirichlet allocation. *Journal of Machine Learning Research*, 3, 993–1022.
- Block, J. H., De Vries, G., Schumann, J. H., & Sandner, P. (2014). Trademarks and venture capital valuation. *Journal of Business Venturing*, 29(4), 525–542.
- Castaldi, C. (2020). All the great things you can do with trademark data: Taking stock and looking ahead. *Strategic Organization*, 18(3), 472–484.
- Chang, J., Gerrish, S., Wang, C., Boyd-Graber, J. L., & Blei, D. M. (2009). Reading tea leaves: How humans interpret topic models. *Advances in Neural Information Processing Systems*, 22, 288–296.
- Dewar, R. D., & Dutton, J. E. (1986). The adoption of radical and incremental innovations: An empirical analysis. *Management Science*, 32(11), 1422–1433.
- Dinlersoz, E. M., Goldschlag, N., Myers, A., & Zolas, N. (2018). An anatomy of U.S. firms seeking trademark registration. NBER Working Paper No. 25038.
- Dinlersoz, E., Goldschlag, N., Yorukoglu, M., & Zolas, N. (2023). On the role of trademarks: From micro evidence to macro outcomes. US Census Bureau Center for Economic Studies Working Paper.
- Fleming, L. (2001). Recombinant uncertainty in technological search. *Management Science*, 47(1), 117–132.
- Flikkema, M., Castaldi, C., de Man, A.-P., & Seip, M. (2019). Trademarks’ relatedness to product and service innovation: A branding strategy approach. *Research Policy*, 48(6), 1340–1353.
- Flikkema, M., Castaldi, C., & de Man, A.-P. (2025). On trademarks and innovation: A retrospective, 10 years later. *Industry and Innovation*, 32(9), 1081–1086.
- Golder, P. N., & Tellis, G. J. (1993). Pioneer advantage: Marketing logic or marketing legend? *Journal of Marketing Research*, 30(2), 158–170.

- Graham, S. J. H., Hancock, G., Marco, A. C., & Myers, A. (2013). The USPTO trademark case files dataset: Descriptions, lessons, and insights. *Journal of Economics & Management Strategy*, 22(4), 669–705.
- Hall, B. H., Jaffe, A. B., & Trajtenberg, M. (2001). The NBER patent citations data file: Lessons, insights and methodological tools. NBER Working Paper No. 8498.
- Henderson, R. M., & Clark, K. B. (1990). Architectural innovation: The reconfiguration of existing product technologies and the failure of established firms. *Administrative Science Quarterly*, 35(1), 9–30.
- Hoberg, G., & Phillips, G. (2016). Text-based network industries and endogenous product differentiation. *Journal of Political Economy*, 124(5), 1423–1465.
- Hsu, G. (2006). Jacks of all trades and masters of none: Audiences’ reactions to spanning genres in feature film production. *Administrative Science Quarterly*, 51(3), 420–450.
- Kalyani, A. (2024). The creativity decline: Evidence from US patents. Federal Reserve Bank of St. Louis Working Paper 2024-008.
- Kelly, B., Papanikolaou, D., Seru, A., & Taddy, M. (2021). Measuring technological innovation over the long run. *American Economic Review: Insights*, 3(3), 303–320.
- Lau, J. H., Newman, D., & Baldwin, T. (2014). Machine reading tea leaves: Automatically evaluating topic coherence and topic model quality. In *Proceedings of the 14th Conference of the European Chapter of the Association for Computational Linguistics* (pp. 530–539).
- Murdock, J., Allen, C., & DeDeo, S. (2017). Exploration and exploitation of Victorian science in Darwin’s reading notebooks. *Cognition*, 159, 117–126.
- PatentsView. (2024). Patent and assignee data, US Patent and Trademark Office. <https://patentsview.org>.
- Seip, M., Castaldi, C., Flikkema, M., & de Man, A.-P. (2018). The timing of trademark application in innovation processes. *Technovation*, 72–73, 34–45.
- Semadeni, M., & Anderson, B. S. (2010). The follower’s dilemma: Innovation and imitation in the professional services industry. *Academy of Management Journal*, 53(5), 1175–1193.
- Silver, E. (2026). An event-dated corpus of US trademark prosecution and a two-sided measure of vocabulary position. Working paper. Code and data: <https://github.com/ericsilver/tm-vocabulary>.
- Taeuscher, K., & Rothe, H. (2021). Gaining legitimacy by being different: Optimal distinctiveness in crowdfunding platforms. *Academy of Management Journal*, 64(1), 149–179.
- Tushman, M. L., & Anderson, P. (1986). Technological discontinuities and organizational environments. *Administrative Science Quarterly*, 31(3), 439–465.
- Uzzi, B., Mukherjee, S., Stringer, M., & Jones, B. (2013). Atypical combinations and scientific impact. *Science*, 342(6157), 468–472.

- Verhoeven, D., Kovacs, A., Marullo, C., Di Minin, A., & Van Looy, B. (2025). More or the same? Radical, disruptive, discontinuous, and breakthrough innovation. *Industrial and Corporate Change*, 34(4), 744–774.
- Willeke, T., Block, J., & Lambrecht, D. (2026). Using trademark data in research. *World Patent Information*, 84, 102431.
- Zhao, E. Y., Fisher, G., Lounsbury, M., & Miller, D. (2017). Optimal distinctiveness: Broadening the interface between institutional theory and strategic management. *Strategic Management Journal*, 38(1), 93–113.
- Zuckerman, E. W. (1999). The categorical imperative: Securities analysts and the illegitimacy discount. *American Journal of Sociology*, 104(5), 1398–1438.